

DSA0405 - Fundamentals of Data Science

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Short Answer Test

①. What is Data Science? Explain its primary Objective.

Data Science is an interdisciplinary field that uses scientific methods, algorithms, and machine learning to extract knowledge and insights from data.

Primary Objective:

To collect, process, analyze and interpret data for better decision-making and predictions.

②. Why is Data Science important in today's digital world? Mention any four reasons.

- * Helps Organizations make data-driven decisions.
- * Improves business efficiency and productivity.
- * Predicts future trends using historical data.
- * Supports automation and AI-based solutions.

③. List any four benefits of data science in business and society.

- * Better decision-making.
- * Improved customer satisfaction.

- * Fraud detection and risk management
- * Healthcare improvements through disease prediction.

④. Mention four real-world applications of Data Science

- * Healthcare (disease diagnosis)
- * Banking (fraud detection)
- * E-Commerce (recommendation Systems)
- * Transportation (traffic and route Optimization)

⑤. What are the different facts of data? Briefly explain each.

- * Structured Data: Organized in rows and columns (e.g., SQL database)
- * Semi-Structured Data: Partially organized with tags (e.g., XML, JSON)
- * Unstructured Data: No predefined format (e.g., images, videos, emails)
- * Metadata: Data that describes other data (e.g., file size, creation data)

6. Differentiate between structured, semi-structured, and unstructured data with suitable examples.

Type	Description	Example
Structured	Organized in tables	student database
Semi-structured	Contains tags but no fixed schema.	XML, JSON
Unstructured	No predefined format.	Images, videos, social media posts.

7. What is Big data? State its key characteristics (5 Vs).

Big Data refers to extremely large and complex datasets that cannot be processed using traditional tools.

5 Vs:

1. Volume

2. Velocity

3. Variety

4. Veracity

5. Value.

8. Explain the Big Data? State its Key ecosystem.

Name any four technologies used in it.

The Big Data ecosystem consists of tools used for collecting, storing, processing, analyzing and

Visualizing large datasets.

Technologies:

- * Hadoop.
- * Apache Spark.
- * Hive.
- * HBase.

9. What are the major phases of the "Data Science process?"

- * Problem Definition.
- * Data Collection.
- * Data Cleaning.
- * Data Exploration.
- * Data Analysis.
- * Model Building.
- * Evaluation.
- * Deployment.

10. What is data retrieval? Mention two common sources from which data can be retrieved.

Data retrieval is the process of obtaining data from storage systems for analysis.

Sources:

1. Databases.
2. APIs (or Web Services).

11. Why is data collection considered an important step in the Data Science process?

- * Provides accurate and relevant data.
- * Improves model performance.
- * Helps make reliable decisions.
- * Reduces errors in analysis.

12. What is data cleansing? List any four common data-cleaning techniques.

Data cleansing is the process of identifying and correcting inaccurate or incomplete data.

Techniques:

- * Remove duplicate records.
- * Handle missing values.
- * Correct Inconsistent data.
- * Remove outliers and errors.

13. What is data Integration? Why is it necessary in Data Science?

Data Integration is the process of combining data from multiple sources into a Unified dataset.

Necessity.

- * Improves data quality.
- * Eliminates redundancy.
- * Provides a Complete view of Information.
- * Enables better analysis and decision-making.

14. Explain data transformation with suitable examples.

Data transformation is the process of converting data into a suitable format for analysis.

Ex: * Converting text categories into numeric values.

- * Standardizing data formats.
- * Scaling numerical data.
- * Normalizing values between 0 and 1.

15. What is data analysis? Mention any four techniques used for data analysis.

Data Analysis is the process of inspecting, cleaning, transforming and interpreting data to discover useful information.

- * Descriptive Analysis
- * Diagnostic Analysis
- * Predictive Analysis
- * Prescriptive Analysis.

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